SONIA REILLY

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OVERVIEW

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Applied mathematician with expertise in numerical methods, uncertainty quantification, Bayesian statistics, and inverse problems. Recent research areas include Bayesian inverse problems governed by PDEs, problems with hyperparameters, and optimal experimental design.

EDUCATION

Courant Institute of Mathematical Sciences, New York University — New York, NY
Ph.D. in Mathematics — *Expected 2027*
Thesis Advisor: Georg Stadler
Supported by Department of Energy Computational Science Graduate Fellowship

Massachusetts Institute of Technology — Cambridge, MA
S.B. in Mathematics with Computer Science — *2021*

RESEARCH EXPERIENCE

Courant Institute of Mathematical Sciences — New York, NY
Graduate Researcher, advised by Georg Stadler — *2022 – present*

Optimal Experimental Design under Hyperparameter Uncertainty

- Investigating how to optimize sensor placement to maximize information gain in Bayesian inverse problems with prior hyperparameters.

Bridging the Gap between INLA and PDE-Governed Inverse Problems

- Synthesizing the differences and striking commonalities between hierarchical Bayesian inverse problems as addressed by the PDE-governed problems community and those addressed by INLA, a widespread method in the geostatistics community.

Amortized Methods for Hierarchical Bayesian Inverse Problems

- Developed scalable deterministic methods of marginalizing hyperparameters in linear Bayesian PDE inverse problems with Gaussian priors.
- Extended state-of-the-art low-rank approximation methods to enable fast computation in problems with nonlinear prior hyperparameters, such as correlation lengths.
- Demonstrated speed and accuracy on a time-dependent 3D advection-diffusion benchmark problem.

MCMC Sampling for Infinite-Dimensional Bayesian Inverse Problems

- Implemented a novel MCMC sampler designed for posterior distributions that emerge from infinite-dimensional Bayesian PDE inverse problems.
- Combined elements of affine invariant methods and the preconditioned Crank Nicolson method to form a gradient-free ensemble sampler that targets high-dimensional, concentrated posteriors.

Lawrence Berkeley National Laboratory — Berkeley, CA
Graduate Research Intern, advised by Jeff Donatelli in James Sethian's group — *Summer 2023*

- Designed a method to incorporate measurement uncertainty into the Multi-Tiered Iterative Phasing (M-TIP) algorithm for recovering the density of a molecule imaged using fluctuation X-ray scattering.
- Successfully retrieved densities from images with added constant and variable Gaussian noise.

Johns Hopkins University Applied Physics Laboratory Laurel, MD
Postgraduate Researcher, advised by Sam Yee *Summer 2021*

- Developed a numerical method for the tomographic inversion problem of recovering atmospheric brightness from satellite data which records integrated brightness values.

Undergraduate Researcher, advised by William Swartz *Sep 2016 - Aug 2017, Summer 2018*

- Designed and implemented a method of extracting high-resolution maps of Earth's outgoing radiation from low-resolution satellite data using spherical harmonics approximation and compressive sensing.
- Applied the method to extract spatial data from RAVAN, NASA's first earth science CubeSat mission.

Massachusetts Institute of Technology Cambridge, MA
Undergraduate Researcher, advised by Jörn Dunkel *Feb 2020 – Dec 2020*

- Developed network topologies optimized to cancel out fluctuations in coupled oscillator systems governed by the swing equation, such as power grid models.
- Categorized the effect of the noise spectrum of the fluctuations on the optimal network structure.
- Quantified the resistance of optimized networks to cascades of line failures in a power grid model.

Undergraduate Researcher, advised by Sai Ravela *Summer 2019*

- Investigated the mathematics of deep learning by modeling the evolution of the weights of neural networks in stochastic gradient descent as a Fokker-Planck system.

PUBLICATIONS

2. **S. Reilly**, G. Stadler. Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-Governed Bayesian Inverse Problems. 2026. Under review at SIAM Journal on Scientific Computing. arXiv preprint: 2607.03355.

1. W. Swartz, S. Lorentz, S. Papadakis, P. Huang, A. Smith, D. Deglau, Y. Yu, **S. Reilly**, N. Reilly, and D. Anderson. "RAVAN: CubeSat Demonstration for Multi-Point Earth Radiation Budget Measurements". *Remote Sensing* 11.7 (Apr. 2019), 796. DOI: 10.3390/rs11070796.

INVITED TALKS AND ORGANIZED SESSIONS

3. SIAM Conference on Computational Science and Engineering, Pittsburgh, PA, February 2027 (upcoming; minisymposium co-organizer)
   Session: Sampling-free Bayesian Inference: Bridging Spatial Statistics and PDE-governed Inverse Problems
   Title: "Amortized Hyperparameter Marginalization for Bayesian PDE-governed Inverse Problems"

2. SIAM Annual Meeting 2026, Cleveland, OH, July 2026
   Session: From Theory to Computation: Modern Methods for PDE Inverse Problems
   Title: "Amortized Low-Rank Approximation for Hyperparameter Marginalization in PDE-governed Bayesian Inverse Problems"

1. SIAM Conference on Computational Science and Engineering, Fort Worth TX, March 2025
   Session: Investigating Inverse Problems using Bayesian Inference: Challenges and Advances
   Title: "Bayesian Inversion of PDE-Governed Problems Using Integrated Nested Laplace Approximations"

CONTRIBUTED TALKS AND POSTERS

5. Numerical Analysis and Scientific Computing RTG Workshop, Rice University, Houston, TX, October 2026 (upcoming)

4. ICERM Workshop on Bayesian Inverse Problems and UQ, Providence, RI, March 2026 (poster)

3. IMSI Workshop on Data Assimilation and Inverse Problems for Digital Twins, Chicago, IL, October 2025 (poster)
2. DOE CSGF Annual Program Review Outgoing Fellow Presentation, Washington D.C., July 2025
1. Mid-Atlantic Numerical Analysis Day, Temple University, Philadelphia, PA, November 2024

AWARDS

SIAM Student Travel Award, SIAM Annual Meeting 2026
SIAM Student Travel Award, SIAM Computational Science and Engineering 2025
Department of Energy Computational Science Graduate Fellowship 2021-2025
NSF Graduate Research Fellowship (Awarded, but declined in favor of DOE CSGF.) 2021
Henry M. MacCracken Fellowship, New York University 2021

TEACHING AND MENTORSHIP

Graduate Teaching Assistant, Courant Institute, NYU

MATH-UA 334: Mathematical Statistics *Spring 2025, 2026 and Fall 2024, 2025, 2026*

- Led two weekly 75 minute recitation sections per semester, each for 30-40 undergraduate juniors and seniors, and held weekly office hours.
- Wrote over 70 pages of LaTeX-typeset problems, solutions, and lecture reviews, led exam review sessions, and engaged with students on the class discussion board.
- Student evaluations: 4.76/5 average across semesters

MATH-UA 253: Linear and Nonlinear Optimization *Spring 2024*

- Led weekly 75 minute recitations and held office hours for a class of 45 undergraduate seniors.
- Developed Julia coding tutorials, wrote example problems and solutions for weekly worksheets, and prepared reviews of lecture material. Graded exams and responded to discussion board questions.
- Delivered a 90-minute lecture as a substitute for the course instructor.
- Student evaluations: 4.75/5

Research Mentor, Courant Institute, NYU

- Anchi Lin, NYU undergraduate student, Summer 2026
  Project: Delayed Acceptance MCMC for Bayesian Inverse Problems with Hyperparameters

SERVICE AND OUTREACH

CAS-GSAS Student Network Mentoring Program, Graduate Mentor 2026
- Mentored an NYU rising senior applying to graduate programs in math.

Courant Association of Math PhD Students, Co-President 2023–2024
- Interfaced with the department, raised funds, and organized visiting students' day.
- Oversaw a revamp of the student-organized website that hosts resources for incoming students.

Courant Diversity, Equity, and Inclusion Reading and Outreach Group, Member 2021–2024
- Led and participated in discussions on diversity and equity in mathematics and graduate education.
- Volunteered at a local high school, assisting in evaluating student math projects.

Association for Women in Math, Courant Organizing Committee Member 2021–2023
- Organized social events for women in the Courant math department, and led an effort to invite more women speakers to department colloquia.

Courant Graduate Student/Postdoc Seminar, Co-organizer 2021–2022

- Invited speakers and organized logistics for a seminar connecting first-year students with potential advisors.

MIT Undergraduate Society of Women in Math, Peer Mentor 2020–2021

- Mentored a younger undergraduate woman in navigating the math department at MIT.

North American Computational Linguistics Open Competition, Organizing Committee 2017–2021

- Helped organize, contributed problems, served on the jury, and hosted the MIT location of this contest for high school students.

SKILLS

Software	Python, Julia, MATLAB, C/C++, parallel computing: CUDA, OpenMP
Languages	Spanish (fluent); German, Italian, French (proficient in reading)